Table II

Effects of Cold-acclimatization upon Activities of Rate-limiting Enzymes within the Liver Lobule of Fed and Fasted Rats [a.]

Ambient Temperature		4°C		24°C		P [b.] for Difference between Periportal and Central Areas of Lobules	P [b.] for Difference between Rats Exposed to 24°C and 4°C	P [b.] for Difference between Fasted and Fed Rats [a.]
Nutrition 12 Hours before Sacrifice		Fed	Fasted	Fed	Fasted			
Enzyme	Area of Lobule	mu moles/ug protein/hour						
Glycogen Synthetase	Periportal	0.299	0.258	0.371	0.256	<0.001	<0.10 [c]	<0.05
	Central	0.245	0.224	0.260	0.215		NS	NS
Glucokinase	Periportal	0.039	0.040	0.143	0.161	<0.001	<0.05	NS
	Central	0.334	0.301	0.509	0.433		<0.05	<0.05
Hexokinase	Periportal	0.382	0.253	0.491	0.228	NS	NS [c]	<0.01
	Central	0.349	0.244	0.382	0.244		NS	<0.05
Phospho-fructokinase	Periportal	0.707	0.819	0.920	0.736	<0.10 [d]	NS	NS [e]
	Central	0.756	0.932	0.949	1.032		<0.05	<0.05
Fructose Diphosphatase	Periportal	12.50	10.09	11.72	8.59	<0.001	<0.10	<0.05
	Central	4.41	5.16	3.76	4.76		NS	NS
Glucose 6-phosphatase	Periportal	68.86	72.25	50.02	62.72	<0.001	<0.05	<0.05
	Central	36.57	46.15	29.87	39.39		<0.05	<0.001
Glutamic-pyruvic transaminase	Periportal	14.52	14.59	14.69	13.41	0.001	0.10 [d]	NS [d]
	Central	4.55	3.97	3.59	3.06		0.10	0.10

a. Animals were either fed or fasted for 12 hours after 1 month exposure to ambient temperatures of 24°C or 4°C.
b. Probability of differences occurring by chance according to analysis of variance; NS, difference nonsignificant.
c. Significant by Student t-test for fed rats. d. Significant by Student t-test for fasted rats.
e. Significant by Student t-test for 24° C-acclimatized rats.

Source: https://www.industrydocuments.ucsf.edu/docs/zlmg0227